Do Now · retrieval

Thinking procedurally: ordering subprocedures

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why does `age>12 && age<18` reject two valid ages for a range 12–18 inclusive?

Q2

What is a subprocedure?

Q3

What is an object's state?

Q4

Why not send a success message before saving a transaction?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Decompose the task — Separate the problem into input validation, checking availability, updating the record and notifying the user.

Core idea 2

Identify dependencies — No booking should be saved until seats are available; no confirmation should be sent before a successful save.

Core idea 3

Make smaller operations reusable — ValidateRequest and SendConfirmation can become independently testable subprocedures.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

decomposition — Split a problem into smaller operations.

subprocedure — Named step used within a larger solution.

dependency — Stage that must precede another.

state update — Change the stored record or value.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

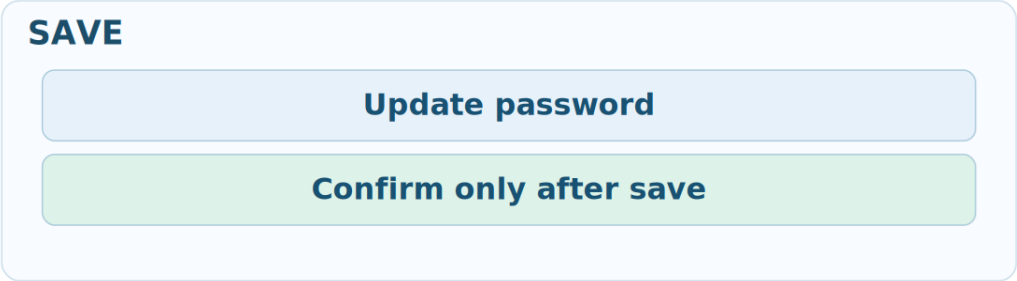
Problem

Put password-reset stages in a safe order.

Follow the worked method

1. Predict first — First identify the account and create a single-use reset token.
2. Model the reasoning — Check the token before accepting a new password.
3. Model the reasoning — Save the change successfully before confirming to the user.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Identify the book and verify availability before trying to record a loan.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Order these steps for ticketing: record purchase, take request, check payment, issue ticket.

Question 2

Design four named subprocedures for booking a classroom and give two key ordering requirements.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A booking code subtracts seats from a local int parameter but never updates the central system. Identify three problems and three remedies.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A booking confirmation can safely be sent before the record is saved.

Use key vocabulary from memory

1. Explain decomposition in your own words.

2. Explain subprocedure in your own words.

3. Explain dependency in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.